AKHILESH B S

Unreal Engine Developer · XR & Simulation · Digital Twins

bsakhileshshetty@gmail.com | +91 7989591934 | LinkedIn: Akhilesh B S | GitHub: BS Akhilesh | Portfolio

PROFESSIONAL SUMMARY

Unreal Engine 5 developer with 2.5+ years building production-grade real-time systems across flight simulation, architectural digital twins, and XR applications. Deep hands-on experience integrating JSBSim physics over UDP, engineering WebSocket and REST pipelines for cross-device control, and shipping optimized VR builds for Meta Quest and Varjo headsets. Comfortable owning systems end-to-end — from C++ and Blueprint architecture to Android APK interfaces and live data integration. Proven in fast-moving teams where simulation accuracy and real-time performance are non-negotiable.

TECHNICAL SKILLS

Core Technologies: Unreal Engine 5, C++, Blueprints, XR (AR/VR/MR), Digital Twin Systems
Simulation & Systems: JSBSim, UDP/TCP Networking, WebSockets, VRPN, REST APIs, JSON, Real-time Data Integration
Procedural & Environment: PCG Framework, Geo-Spatial Data, Dynamic Weather Systems
Visualization & UI: UMG, Pixel Streaming, Android APK UI Systems, Cross-device Control Interfaces, Lumen, Nanite, Sequencer, 4K Rendering
Platforms & Tools: Meta Quest, Varjo, Android, Blender, Git/GitHub, Python, Unity

WORK EXPERIENCE

Unreal Engine Developer | Credvest · **Full-time** | *Oct 2025 – Present*

- Architected a real-time architectural Digital Twin platform in UE5, enabling clients to visualize and interact with live property inventory synced from Google Sheets and CSV data sources.
- Engineered a WebSocket-based remote control system allowing the UE5 application to be operated from tablets and Android APK interfaces, reducing on-site navigation friction for sales demos.
- Built weather-driven digital twin environments using live weather APIs, dynamically updating sky, lighting, and environmental conditions in real time.
- Developed a structured analytics system tracking user interactions, session activity, and feature usage with JSON-based data storage providing actionable insight into demo engagement.
- Delivered optimized Android and VR builds through performance profiling and runtime optimization, maintaining stable frame rates across target devices.
- Authored scalable Blueprint and C++ systems adopted across multiple production workflows, cutting integration time for new features.

Junior Unreal Engine Developer | LTX-LaunchTrax · **Full-time** | *Sep 2024 – Oct 2025*

- Engineered a full-featured flight simulation system integrating JSBSim physics over UDP, supporting realistic aircraft dynamics, carrier-based operations, multiplayer missions, and air-to-air missile behavior.
- Built large-scale procedural environments using UE5 PCG with geo-spatial latitude/longitude data, enabling dynamic terrain generation and scalable environment automation across mission scenarios.
- Implemented a VRPN-based synchronization pipeline between Unreal Engine and Unity, enabling real-time distributed simulation across two engines simultaneously.
- Rigged and animated multiple aircraft models; developed cockpit HUD/UI systems used in live dogfight and carrier landing simulation sessions.
- Delivered optimized VR simulations for Meta Quest and Varjo headsets with stable real-time networking under active multiplayer load.

Game Developer | LTX - LaunchTrax · Internship | Jul 2024 – Aug 2024

- Developed helicopter HUD systems and flight simulation workflows using JSBSim, reaching production-level output within the first two weeks.
- Converted JSBSim into a reusable, self-contained Unreal Engine plugin, eliminating repetitive integration setup for all future simulation projects on the team.

KEY PROJECTS

Flight Simulation: JSBSim Integration

Built a complete real-time flight simulation system in UE5 covering JSBSim physics over UDP, aircraft rigging and animation, cockpit HUD systems, and multiplayer mission workflows. Supported carrier landing and dogfight scenarios with stable frame rates across Meta Quest and Varjo VR headsets.

Architectural Digital Twin Application

Developed a real-time architectural visualization platform for property sales, integrating live inventory data via Google Sheets and CSV. Included a WebSocket-driven remote-control layer operable from Android devices and tablets, with dynamic weather and lighting driven by live API data.

XR Hospital Interaction System

Designed immersive XR workflows for a hospital environment, including patient monitoring dashboards, real-time data visualization, and interactive object systems targeting clinical training and simulation use cases.

Real-Time Communication & Control System

Implemented a UDP and WebSocket communication pipeline with structured JSON data exchange, enabling real-time application control from Android APK interfaces and external devices. Achieved low-latency, bidirectional control with reliable state synchronization.

Runtime Modeling Tool

Developed a runtime 3D editing tool inside UE5 supporting undo/redo stacks, transform controls (translate, rotate, scale), object duplication, and save/load persistence delivering a Blender-like editing workflow directly inside a packaged application without editor access.

EDUCATION

B.Tech in Mechanical Engineering · SV College of Engineering and Technology · 2019–2022 · 85%

Diploma in Mechanical Engineering · Kuppam Engineering College · 2015–2019 · 68%

CERTIFICATIONS & TRAINING

Artificial Intelligence in Unreal Engine 5 — *Packt / Coursera · 4-Week Certification*
AI systems in UE5 · AI Tools and Features · Practical AI Development

AR/VR Experiences — *Ajna Creator · 6-Month Intensive*
XR Interaction · Simulation Systems · Digital Twin Development · 3D Environment Creation

Full Stack Developer — *JSpiders · 6-Month Training*
Java · HTML · CSS · JavaScript · React · SQL